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Cambridge Ordinary Level

MARK SCHEME for the October/November 2014 series

5054 PHYSICS

5054/22 Paper 2 (Theory), maximum raw mark 75

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Section A

1	(a)	limit of proportionality (<u>not</u> breaking point)	B1	
	(b) (i)	8.5 cm **cao**	B1	
	(ii)	7.1 – 7.3 cm	B1	
	(c)	$4.0 \times (2.7 / 7.2)$ **or** $5.0 \times (2.7 / 9.0)$ **or** 1.5 (N) **or** read from graph **or** 11.2 (cm)	C1	
		0.148 – 0.152 kg **or** 148 – 152 g	A1	**[5]**
2	(a) (i)	Fd **or** 2.5×0.18	C1	
		0.45 N m	A1	
	(ii)	force not applied at right angles to the tap	B1	
	(b)	long(er) distance needs small(er) force (for same moment) **or** inversely related/proportional	B1	**[4]**
3	(a)	$V_1 = p_2V_2 / p_1$ **or** $p \propto 1 / V$	B1	
		$1.0 \times 10^5 \times (1.8 / 2.0) \times 10^7$	C1	
		9.0×10^{-3} m^3 **or** 9000 cm^3	A1	
	(b) (i)	$(\rho =) m / V$ **or** $(0.30 / 9.0) \times 10^{-3}$	C1	
		33(.3333) kg / m^3 **or** 0.033(3333) g / cm^3	A1	
	(ii)	helium mass / weight small (fraction of total / mass of air included) **or** this includes the weight of the cylinder	B1	**[6]**
4	(a) (i)	heat gained from burning fuel / combustion **or** friction between moving parts / with air / road **or** from (radiation of) Sun	B1	
	(ii)	heat lost to air/surroundings **or** by convection (currents) **or** exhaust/hot gases/fumes **or** from exhaust **or** heat emitted (by hot car) **or** by radiation	B1	
	(b)	at start chemical energy decreases **or** at start chemical energy to	B1	
		gravitational / potential energy (of car) increases **or** at end of process	B1	
		kinetic energy (of car **or** air) increases	B1	**[5]**

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5	**(a)**	downward curve of correct curvature from marked 90 °C		B1	
		horizontal line at marked 58 °C		B1	
		downward (asymptotic) curve of correct curvature to marked 23 °C		B1	
	(b)	H marked halfway (by eye) along an intermediate horizontal line		B1	
	(c)	(Q =) mL **or** 45 × 220		C1	
		9900 J		A1	**[6]**
6	**(a)**	(the molecules) move faster **or** have more **kinetic** energy **or** accelerate **ignore** vibrate faster		B1	
	(b)	**(i)** faster / energetic molecules escape		B1	
		average speed decreases **or** slower molecules remain		B1	
		temperature depends on average KE **or** heat taken from runner **OR** liquid becomes gas / vapour latent heat needed or bonds broken heat taken from runner		B1	
		(ii) water vapour blown away **or** surrounding air less humid		B1	**[5]**
7	**(a)**	**(i)** lasts longer **or** one cell can be replaced without switching off the circuit **or** less (internal) resistance **or** if one fails the others still work		B1	
		(ii) 1.5 V		B1	
	(b)	**(i)** (R =) V / I **or** 1.5 / 0.075		C1	
		20 (Ω) **or** 1.5 / (0.075 – 6.0)		C1	
		14 Ω		A1	
		(ii) decreases		B1	
		resistance of wire increases		B1	**[7]**
8	**(a)**	one label correct and not contradicted		C1	
		C, 1S and 1B all correct and clear and none contradicted		A1	
	(b)	any **three** from: magnetic field (between poles) (coil / wire) cuts field / flux **or** field / flux cuts (coil / wire) **or** field / flux changes (electromagnetic) induction brushes rub against / in contact with rings		B3	
	(c)	(half) distance across screen **or** count divisions of / measure wavelength **or** the wavelength corresponds to one rotation		B1	
		half distance multiplied by time base setting		B1	**[7]**

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Section B

9 (a) changing speed / velocity — C1
change in speed / velocity / time constant **or** (v–u) / t constant **or** constant / equal rate of change of speed / velocity — A1 **[2]**

(b) (a vector quantity has) direction — B1 **[1]**

(c) (i) **1.** X between $t \geq 0$ and $t < 10$ s — B1
2. Y between $t > 20$ s and $t < 30$ s — B1
3. Z between $t > 10$ s and $t < 20$ s **or** between $t > 30$ s and $t < 40$ s — B1

(ii) **1. two** speed values from graph between 15 and 35 s (±1 mm) — C1
two corresponding time values from graph between 15 and 35 s (±1 mm) **or** (a =)$\Delta v / t$ — C1
500 m / s^2 — A1
2. (F =) ma **or** 8.4 × 500 — C1
4200 N — A1

(iii) **1.** arrow labelled F perpendicular to surface of Earth — B1
arrow labelled R opposite to direction of travel (by eye) from rock — B1
2. speed changes **or** density / pressure of air changes **or** cross-sectional area (of rock) changes — B1

(iv) it hits the ground / surface of the earth **or** stops **or** speed is zero — B1 **[12]**

[Total: 15]

10 (a) 3.0×10^8 m / s — B1 **[1]**

(b) (i) **1.** decreases **cao** — B1
2. no change **cao** — B1
3. decreases **cao** — B1

(ii) **1.** i correctly marked (to normal) — B1
2. r correctly marked (to normal) — B1 **[5]**

(c) (i) sin i / sin r = n **or** sin i / sin r = 1.5 — C1
sin 89 / sin r = 1.5 **or** sin 89 / 1.5 **or** 0.67(0.666565) — C1
42° **or** 41.8025° — A1

(ii) i equal to / close to 90° **and** r less than 45° — sin i / sin 45 = 1.5 — $\sin^{-1}(1/n) / \sin^{-1}(1/1.5)$ **and** 41.8° — B1
or **or**
i never bigger than 89° / 90° — sin i >1 — r not be more than c — B1 **[5]**

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(d) (i) (sin) $i = 0$ **or** ray enters directly / along normal / perpendicular **or** wavefront / light hits surface all together B1
(sin) $r = 0$ **or** no refraction all slows down together B1

(ii) correct reflection at bottom surface (by eye) M1
second correct reflection at top and no refraction at either point A1 **[4]**

[Total: 15]

11 (a) same element **or** same number of protons / atomic number B1
different / particular number of neutrons **or** nucleons B1 **[2]**

(b) (i) 38 **cao** B1

(ii) 52 **cao** B1 **[2]**

(c) $^{90}_{39}(\mathrm{Y})$ **or** $^{90}(\mathrm{Y})$ and $^{0}(\beta)$ B1
$^{0}_{-1}(\beta)$ $_{39}(\mathrm{Y})$ and $_{-1}(\beta)$ B1 **[2]**

(d) (i) 87 / 29 **or** 3 (half-lives) **or** $6.0 \times 10^8 / 8$ C1
7.5×10^7 A1

(ii) any detector B1
corresponding detection method B1

detector	film	(solid-state) detector	GM-tube	ionisation chamber	scintillation counter	cloud chamber
detection	fogged	count/ reading	count/ reading	count/ reading	count/ reading	track seen

no reduction with paper **or** (use of) electric/magnetic field **or** describe pattern of track M1

complete reduction with aluminium/lead **or** correct deflection of track in electric/magnetic field **or** no other track A1

(iii) **1.** unpredictable **or** not be known in advance **or** no set time between emissions **or** fluctuates **or** not fixed **or** counts obtained varies B1
2. any **two** from:
direction / in space
time
which nucleus decays B2 **[9]**

[Total: 15]

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